

Exact Precision and Algebraic Rigidity of Omnific Continued Fractions

Full Hahn-field realization, valuation ideals,
periodic codes, and surcomplex fibers

AI-assisted research draft

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Abstract

An ordinary infinite continued fraction with omnific integer digits does not usually determine one surreal number. Its indeterminacy was already studied by Roggeman. We give a valuation-theoretic refinement and use it to derive precise realization and algebraicity criteria. If q_n are the continuant denominators and $d_n = \deg q_n$, the complete fiber of a digit sequence is exactly a translate of the ideal of errors ϵ satisfying $v(\epsilon) > 2d_n$ for every ordinary n . A uniform two-step buffer estimate makes this description independent of the chosen realization.

Every admissible sequence in a full real-coefficient Hahn field has a realization in that same field; it has exactly one precisely when the cumulative denominator degrees are cofinal in the value group. We classify when the precision ideal is prime and give an exact two-condition criterion for the fiber to contain at most one element algebraic over its digit field. For eventually periodic codes an algebraic realization always exists, so this becomes a sharp one-versus-countably-many dichotomy. Each full surreal fiber nevertheless contains algebraically independent families of every set cardinality over every prescribed set-sized coefficient field. Coordinatewise surcomplex fibers and their changes under fractional linear maps complete the picture. The proofs are written in full, with finite symbolic checks and an explicit research agenda.

Status and priority. This is an unrefereed, AI-assisted research draft, not a certified resolution of a named longstanding open problem. Basic surreal continued-fraction nonuniqueness is *not* claimed as new. The proposed contributions are the sharpened precision, workspace, and algebraic-fiber classifications. Their novelty has not been independently certified. The accompanying finite tests are not a Lean verification or a proof of the infinite-support assertions.

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1 The question and the contribution

Over the real numbers, an infinite regular continued-fraction expansion is a representation of a single irrational number. Over the surreal numbers, there are two different questions: which points produce a specified digit sequence, and which point should be selected as its distinguished value? The first is an arithmetic question about the floor algorithm. The second requires an additional selection principle. Confusing them makes a periodic code look more rigid than it is.

This article studies the first question, with digits in Conway's omnific integer ring $\mathbf{Oz}$. The sequence length is exactly ω : all indices n, k, m used in the continued-fraction algorithm are *ordinary finite* natural numbers. Neither a limit-stage continuation of the algorithm nor a Gaussian nearest-integer algorithm is silently included.

1.1 What was already known

Roggeman's *Continued fractions and surreal numbers* [3] already develops the omnific floor algorithm, associated classes of points with the same infinite code, and periodic fractions. In particular, his Section 3A characterizes associated classes by convergent inequalities; Section 3H studies their size and structure; Section 3I introduces a generalized continuation. Thus nonuniqueness and a proper-class fiber are antecedents, not discoveries of the present article. The relevant scanned pages were inspected directly.

The continuant identities and the real periodicity theorem are classical; [4, 5] give accessible treatments, with [5] providing a formalized real theory. Our finite identities are rederived because the infinite digits and non-Archimedean inequalities require careful hypotheses. Conway's and Gonshor's normal-form and real-closedness theorems are the surreal foundations used here [1, 2].

The repository [6] supplies the setting and an implementation roadmap. The inspected root README and report catalogue include the actual omnific floor, degree, and normal-form infrastructure. No continued-fraction report was identified in that catalogue. This was a focused comparison, not a declaration-by-declaration or whole-history audit. No unreviewed advanced repository theorem is required below.

1.2 The central classification

Let $\mathbf{a} = (a_0, a_1, \dots)$, where $a_0 \in \mathbf{Oz}$ and $a_n \in \mathbf{Oz}$, $a_n \geq 1$ for $n \geq 1$. Write $\mathcal{F}(\mathbf{a})$ for all surreals whose nonterminating floor algorithm produces this code. Define the usual continuant denominators by

$$q_{-1} = 0, \quad q_0 = 1, \quad q_n = a_n q_{n-1} + q_{n-2} \quad (n \geq 1),$$

and put

$$d_0 = 0, \quad d_n = \sum_{j=1}^n \deg a_j = \deg q_n. \quad (1.1)$$

Here $\deg$ denotes the greatest *growth exponent* in Conway normal form; the natural valuation has the opposite sign.

For every $x \in \mathcal{F}(\mathbf{a})$,

$$\mathcal{F}(\mathbf{a}) = x + I_{\mathbf{a}}, \quad I_{\mathbf{a}} = \{0\} \cup \{\epsilon \neq 0 : v(\epsilon) > 2d_n \text{ for every } n\}.$$

The sequence records a coset of an explicit valuation ideal, rather than a single unrestricted surreal number. The ideal depends only on the degrees of the digits after a_0 , not on their other coefficients.

The exact formula is proved in Theorem 4.4, using a buffer from both cylinder endpoints. It refines the earlier associated-class viewpoint into a form usable for Hahn-field descent and algebraic rigidity.

1.3 Consequences that go beyond basic nonuniqueness

Theorem 5.1 proves existence in every *full* $\mathbb{R}$ -coefficient Hahn field containing the digits, not merely existence somewhere in **No**. Its uniqueness criterion is exactly cofinality of (d_n) in the value group. The existence proof is not a completeness assertion about arbitrary non-Archimedean ordered fields.

For the countable digit field $K_a = \mathbb{Q}(a_0, a_1, \dots)$, set $D_0 = \deg a_0$ when $a_0 \neq 0$ and $D_0 = 0$ otherwise. Two simple conditions control algebraic rigidity:

$$\text{for every } n \text{ there is } m \text{ with } d_m \geq 2d_n, \quad (\text{P})$$

$$\text{there is } n \text{ with } D_0 \leq 2d_n. \quad (\text{V})$$

When both hold, the fiber has at most one element algebraic over K_a . If either fails and one algebraic realization exists, there are countably infinitely many, all obtainable in the same finite field extension by rational translations. This is Theorem 7.3. For eventually periodic codes, existence of an algebraic realization turns this conditional statement into an exact dichotomy (Theorem 8.2).

The prime-ideal criterion, the algebraic dichotomy, and the universal Hahn-workspace realization theorem are the main proposed research contributions. We do not assert that a literature search establishes their priority. Their mathematical content is specified by the statements and proofs, not by a label such as “breakthrough.”

2 Surreal and Hahn conventions

2.1 Normal forms, degree, and valuation

We use the following standard normal-form facts as inputs [2]. Every nonzero surreal has a unique expansion

$$x = \sum_{\gamma \in S} r_\gamma \omega^\gamma, \quad r_\gamma \in \mathbb{R} \setminus \{0\}, \quad (2.1)$$

where S is a reverse well-ordered *set* of surreals. Every such normal-form series exists. Its greatest exponent and leading coefficient determine its growth and sign. Addition and multiplication are the Hahn operations, and $\omega^{\gamma+\delta} = \omega^\gamma \omega^\delta$. We write

$$\deg x = \max S, \quad v(x) = -\deg x, \quad v(0) = +\infty.$$

In particular $v(xy) = v(x) + v(y)$ for nonzero x, y , and $v(x+y) \geq \min\{v(x), v(y)\}$, with equality when the two valuations differ. For positive x, y , leading coefficients do not cancel, so

$$\deg(x+y) = \max\{\deg x, \deg y\}. \quad (2.2)$$

The symbol $\deg 0$ is not used in these identities. The convention $D_0 = 0$ for a zero first digit is a separate bookkeeping definition.

Let $\mathcal{O}$ be the ring of finite surreals and $\mathfrak{m}$ its infinitesimal ideal:

$$\mathcal{O} = \{0\} \cup \{x : v(x) \geq 0\}, \quad \mathfrak{m} = \{0\} \cup \{x : v(x) > 0\}.$$

The residue field is $\mathbb{R}$. For $b > 0$ we use

$$|e| < b \iff |e| < b/m \text{ for every ordinary integer } m \geq 1. \quad (2.3)$$

For nonzero e , this is equivalent to $v(e) > v(b)$. Indeed, a leading valuation equality makes $|e|/b$ have a positive real leading coefficient, which cannot be smaller than every positive real reciprocal.

2.2 The omnific floor

Definition 2.1. The omnific ring $\mathbf{Oz}$ consists of the normal forms with growth exponents in $\mathbf{No}_{\geq 0}$ and with ordinary integer coefficient at exponent zero. The purely infinite coefficients may be arbitrary real numbers.

A nonzero omnific integer has nonnegative degree. An omnific integer of degree zero is an ordinary nonzero integer. The ring is discretely ordered: its least positive element is 1. These observations follow immediately from the leading term and the integer constant coefficient.

Lemma 2.2 (Exact floor, including the boundary correction). *Each $x \in \mathbf{No}$ has a unique $a \in \mathbf{Oz}$ with $a \leq x < a + 1$. Write $x = x_{>0} + r + e$, where $x_{>0}$ is its positive-growth part, $r \in \mathbb{R}$, and $e \in \mathfrak{m}$. If $r \notin \mathbb{Z}$, then $a = x_{>0} + \lfloor r \rfloor$. If $r \in \mathbb{Z}$, then*

$$a = \begin{cases} x_{>0} + r, & e \geq 0, \\ x_{>0} + r - 1, & e < 0. \end{cases}$$

Proof. In the nonintegral case the real fractional part of r is strictly between zero and one, and adding an infinitesimal preserves both inequalities. In the integral case the displayed correction gives the same bounds. For uniqueness, two distinct omnific integers differ in absolute value by at least one and therefore cannot both lie in $(x - 1, x]$. $\square$

We denote this floor by $\lfloor x \rfloor_{\mathbf{Oz}}$. In particular, if $0 < e \in \mathfrak{m}$, then $\lfloor -e \rfloor_{\mathbf{Oz}} = -1$, not zero. The algorithm below needs the actual floor, not constant-term rounding without this correction.

2.3 Full Hahn fields and class size

For a set-sized additive subgroup $G \subseteq \mathbf{No}$, let

$$K_G = \mathbb{R}((\omega^G)) = \{x \in \mathbf{No} : \text{supp}(x) \subseteq G\}, \quad \mathbf{Oz}_G = K_G \cap \mathbf{Oz}. \quad (2.4)$$

The support in this formula is the set of growth exponents. By the normal form theorem K_G is a set-sized ordered field, with value group G . It is the *full* Hahn field: arbitrary allowed well-ordered valuation supports are admitted, not only finite or grid-based supports. Formula (2.4) includes $G = \{0\}$, when $K_G = \mathbb{R}$.

The floor in Lemma 2.2 preserves K_G , since it deletes terms and changes only the constant coefficient. Consequently the continued fraction of any element of K_G , when defined, has digits in $\mathbf{Oz}_G$.

Class statements can be read in a standard conservative class framework, or as assertions about the definable class of surreal sign sequences over ZFC. Every support, sequence, field generated by specified data, and cut used in a construction below is a set. No large-cardinal assumption, class-length recursion for this algorithm, or proper-class choice of a transcendence basis is used.

3 Finite continuants and open cylinders

Definition 3.1 (Digits and complete quotients). For $x_0 = x$, define recursively

$$a_n = \lfloor x_n \rfloor_{\mathbf{Oz}}, \quad x_{n+1} = \frac{1}{x_n - a_n}$$

provided $x_n \neq a_n$. A sequence $\mathbf{a} = (a_n)_{n \geq 0}$ is *admissible* if $a_0 \in \mathbf{Oz}$ and $a_n \in \mathbf{Oz}$, $a_n \geq 1$ for every $n \geq 1$. Its fiber $\mathcal{F}(\mathbf{a})$ consists of inputs for which all complete quotients exist and the digit sequence is exactly $\mathbf{a}$.

The remainder belongs to $(0, 1)$ whenever the algorithm continues. Hence $x_{n+1} > 1$ and its floor is at least one. Finite terminating expansions are not included in an infinite fiber.

For an admissible code define, including the initial indices,

$$\begin{aligned} p_{-2} &= 0, & p_{-1} &= 1, & p_n &= a_n p_{n-1} + p_{n-2}, \\ q_{-2} &= 1, & q_{-1} &= 0, & q_n &= a_n q_{n-1} + q_{n-2}. \end{aligned} \quad (3.1)$$

Then $p_0 = a_0$ and $q_0 = 1$. All entries belong to $\mathbf{Oz}$.

Lemma 3.2 (Matrix identities). *For every $n \geq 0$,*

$$\prod_{j=0}^n \begin{pmatrix} a_j & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} p_n & p_{n-1} \\ q_n & q_{n-1} \end{pmatrix}, \quad (3.2)$$

$$p_n q_{n-1} - p_{n-1} q_n = (-1)^{n+1}. \quad (3.3)$$

Moreover $q_n > 0$ and $q_n \geq q_{n-1}$, and, for the Fibonacci numbers $F_1 = F_2 = 1$,

$$q_{n+k} \geq F_{k+1} q_n \quad (n, k \geq 0). \quad (3.4)$$

Proof. Multiplication of the last matrix gives the recursions in (3.1); each factor has determinant -1 . For the positive and monotonic assertions start with $q_{-1} = 0, q_0 = 1$ and use $a_j \geq 1$. The recurrence gives $q_{j+2} \geq q_{j+1} + q_j$. Induction on k , starting with $k = 0, 1$, proves (3.4). No Archimedean hypothesis enters. $\square$

Define

$$\Phi_n(t) = \frac{p_n t + p_{n-1}}{q_n t + q_{n-1}}, \quad r_n = \frac{p_n}{q_n}, \quad s_n = \frac{p_n + p_{n-1}}{q_n + q_{n-1}}. \quad (3.5)$$

The map Φ_n is the finite composition $a_0 + 1/(a_1 + 1/(\cdots + 1/(a_n + 1/t)))$.

Proposition 3.3 (Cylinder description). *The points having the first $n + 1$ digits $a_0, \dots, a_n$ and a further complete quotient form the open interval*

$$C_n = \Phi_n((1, \infty)) = (\min(r_n, s_n), \max(r_n, s_n)).$$

These intervals are nested, and

$$|C_n| = \frac{1}{q_n(q_n + q_{n-1})} \leq \frac{1}{q_n^2}. \quad (3.6)$$

If $x = \Phi_n(t)$ with $t > 1$, its distances to the two endpoints are

$$|x - r_n| = \frac{1}{q_n(q_n t + q_{n-1})}, \quad (3.7)$$

$$|x - s_n| = \frac{t - 1}{(q_n + q_{n-1})(q_n t + q_{n-1})}. \quad (3.8)$$

Finally,

$$\mathcal{F}(\mathbf{a}) = \bigcap_{n \geq 0} C_n. \quad (3.9)$$

Proof. Each map $u \mapsto a + 1/u$ takes $u > 1$ into $(a, a + 1)$, so its floor is a . Iterating proves the digit interpretation and nesting. Subtracting two fractional linear expressions and using (3.3) proves (3.6)–(3.8). The determinant also shows strict monotonicity; direct solution for t shows that the image is the entire open interval between r_n and s_n . Membership in every finite cylinder is precisely indefinite continuation with the prescribed digits. $\square$

Corollary 3.4 (Realization in the full surreal field). *Every admissible code has a realization in **No**. In fact, let L and R be the sets of lower and upper endpoints of all the C_n . Then the Conway cut $\{L \mid R\}$ belongs to $\mathcal{F}(a)$.*

Proof. The sets are countable and every member of L is strictly less than every member of R : compare both with a point of a sufficiently late nonempty nested interval. The standard cut property supplies a surreal strictly between the two sets. It lies in every C_n . $\square$

The cut in this corollary is a simplicity-based choice. The proof does not interpret it as an order-topological limit, and later periodic fixed-point choices will not be identified with it without a separate argument.

4 An exact valuation description of every fiber

4.1 Why both endpoints matter

An upper error bound $|x - r_n| < q_n^{-2}$ does not by itself show that a perturbation preserves the digits. One also needs a lower bound on the distance to the *other* boundary. The case of a next digit equal to one is the delicate boundary case.

Lemma 4.1 (Uniform two-step buffer). *For $x \in \mathcal{F}(a)$ and every $n \geq 0$, the distance from x to either endpoint of C_n is strictly greater than*

$$\frac{1}{4q_{n+2}^2}. \quad (4.1)$$

Proof. Put $t = x_{n+1} = a + 1/u$ and $u = x_{n+2} = b + 1/w$, where $a = a_{n+1}$, $b = a_{n+2}$, and $w > 1$. Thus $a < t < a + 1$ and $b < u < b + 1$. Since $q_n t + q_{n-1} < q_{n+1} + q_n \leq 2q_{n+1}$, formula (3.7) gives

$$|x - r_n| > \frac{1}{2q_n q_{n+1}} \geq \frac{1}{2q_{n+1}^2}.$$

For the other endpoint first suppose $a > 1$. Discreteness of **Oz** implies $a \geq 2$, so $t - 1 > 1$. In (3.8), both denominator factors are at most $2q_{n+1}$, and the second is strictly smaller. Consequently $|x - s_n| > 1/(4q_{n+1}^2)$.

If $a = 1$, then $q_{n+1} = q_n + q_{n-1}$ and $t - 1 = 1/u$. Substitution in (3.8) gives the sharper identity

$$|x - s_n| = \frac{1}{q_{n+1}(q_{n+1}u + q_n)} > \frac{1}{q_{n+1}(q_{n+2} + q_{n+1})} \geq \frac{1}{2q_{n+2}^2}.$$

Monotonicity of the denominators now gives (4.1) in all cases. $\square$

Lemma 4.2 (Degree and exact convergent error). *For every $n \geq 0$,*

$$\deg q_n = d_n = \sum_{j=1}^n \deg a_j. \quad (4.2)$$

For each $x \in \mathcal{F}(a)$,

$$v(x - r_n) = d_n + d_{n+1}. \quad (4.3)$$

Proof. The denominator degree formula starts at $q_0 = 1$. In $a_n q_{n-1} + q_{n-2}$ both summands are nonnegative and the first has degree $\deg a_n + d_{n-1} \geq d_{n-2}$. Leading coefficients do not cancel, so induction proves (4.2).

In (3.7), the complete quotient satisfies $a_{n+1} < t < a_{n+1} + 1$, whence $\deg t = \deg a_{n+1}$. The positive expression $q_n t + q_{n-1}$ therefore has degree $d_n + \deg a_{n+1} = d_{n+1}$. Taking the valuation of the reciprocal in (3.7) proves (4.3). $\square$

Definition 4.3 (Precision ideal). For an admissible code let

$$I_a = \{0\} \cup \{e \in \mathbf{No}^\times : v(e) > 2d_n \text{ for all } n \geq 0\}. \quad (4.4)$$

Equivalently, $|e| < q_n^{-2}$ for every n .

Theorem 4.4 (Exact fiber). For every admissible code a and every $x \in \mathcal{F}(a)$,

$$\boxed{\mathcal{F}(a) = x + I_a}. \quad (4.5)$$

The group I_a is a convex $\mathbb{R}$ -linear additive subgroup of $\mathbf{No}$ and a proper ideal of $\mathcal{O}$, contained in $\mathfrak{m}$.

Proof. Suppose $x, y \in \mathcal{F}(a)$. For fixed n and every $k \geq 0$, both points belong to C_{n+k} . Thus by (3.6) and (3.4),

$$|x - y| < q_{n+k}^{-2} \leq F_{k+1}^{-2} q_n^{-2}.$$

The ordinary Fibonacci numbers are unbounded in the ordinary integers. For every ordinary m we may choose k with $F_{k+1}^2 > m$. It follows that $|x - y| < q_n^{-2}$. Since n was arbitrary, $y - x \in I_a$.

Conversely, take $e \in I_a$. For each n , $|e| < q_n^{-2}$, so in particular $|e| < 1/(4q_{n+2}^2)$. The buffer lemma ensures $x + e \in C_n$. This holds for all n , proving $x + e \in \mathcal{F}(a)$.

The valuation inequality proves additive closure, including cancellation. Multiplication by a nonzero real does not change valuation. Multiplication by an element of $\mathcal{O}$ can only increase valuation. Convexity follows either from (2.3) or from the valuation order. Finally $d_0 = 0$ implies $I_a \subseteq \mathfrak{m}$, so $1 \notin I_a$. $\square$

Corollary 4.5 (No unique unrestricted surreal realization). Every admissible infinite code has a nonzero precision ideal and a proper-class fiber in $\mathbf{No}$.

Proof. The set $\{2d_n : n \geq 0\}$ has a strict surreal upper bound γ . Then $\omega^{-\gamma} \in I_a$. For arbitrarily large positive surreal $\eta > \gamma$, the distinct monomials $\omega^{-\eta}$ also lie in the ideal. These cannot form a set: their valuations would give a set unbounded above in $\mathbf{No}$, contrary to the cut property. $\square$

The basic proper-class conclusion is consistent with, and already anticipated by, the older associated-class theory. The useful refinement here is its exact cut of allowed valuations.

4.2 What an ordinary sequence does not converge to

Proposition 4.6 (Order-topological obstruction). A net indexed by a set that converges in the order topology of $\mathbf{No}$ is eventually equal to its limit. In particular, the sequence of continuants r_n of an admissible infinite code does not converge in that topology.

Proof. Suppose $(z_j)_{j \in J}$ converges to z , where J is a set. The nonzero positive distances $|z_j - z|$ form a set S . If S is nonempty, the cut property gives $0 < \delta < s$ for all $s \in S$. Eventual membership in $(z - \delta, z + \delta)$ therefore means eventual equality to z . The continuants are never eventually constant, since the determinant identity makes consecutive ones distinct. $\square$

This elementary class-size obstruction is not presented as a new topology theorem. It explains why cylinder realization, coefficientwise Hahn behavior, and order convergence must be kept separate.

5 Realization inside a prescribed full Hahn field

We now answer a stronger question than Corollary 3.4: when the digits lie in a given full Hahn field, can the realization be kept inside it? For an arbitrary ordered subfield of $\mathbf{No}$ there is no reason for the answer to be affirmative. For K_G it always is.

Theorem 5.1 (Hahn realization and exact uniqueness). *Let $G \subseteq \mathbf{No}$ be a set-sized additive subgroup and let $\mathbf{a}$ be an admissible code in $\mathbf{Oz}_G$. Then*

(i) $\mathcal{F}(\mathbf{a}) \cap K_G \neq \emptyset$;

(ii) for $x \in \mathcal{F}(\mathbf{a}) \cap K_G$,

$$\mathcal{F}(\mathbf{a}) \cap K_G = x + (I_{\mathbf{a}} \cap K_G);$$

(iii) *this fiber is a singleton if and only if $(2d_n)$ is cofinal in $G_{\geq 0}$, in the sense that every $g \in G_{\geq 0}$ satisfies $g \leq 2d_n$ for some n .*

No divisibility assumption on G is needed.

Proof. Only existence requires a new argument. All continuants r_n are already in K_G . Split the proof according to the degree sequence.

Case 1: d_n is eventually constant. Each increment $d_n - d_{n-1} = \deg a_n$ is nonnegative. Thus all sufficiently late digits have degree zero and are positive ordinary integers. The corresponding ordinary real infinite continued fraction has an irrational real value $\rho > 1$ [4]. Apply the finite prefix map Φ_{N-1} to ρ , where the ordinary tail starts at a_N and $N \geq 1$. This produces a point of K_G with the required code. The denominator is positive, and all tail remainders are nonzero by the ordinary irrationality theorem.

Case 2: d_n is not eventually constant. Choose any $x \in \mathcal{F}(\mathbf{a})$ in $\mathbf{No}$. Delete from its normal form the terms whose growth exponents are outside G , and call the resulting subseries $\pi_G(x)$. A subseries of a reverse well-ordered support is allowed, so $\pi_G(x) \in K_G$.

Fix n . There is k with $d_k > d_n$. By (4.3),

$$v(x - r_k) = d_k + d_{k+1} > 2d_n.$$

Since r_k is supported in G , every term of x outside G is also a term of $x - r_k$. Every deleted term therefore has valuation greater than $2d_n$. This is true for every n , and if the deleted subseries is nonzero its leading term is one of those terms. Hence

$$x - \pi_G(x) \in I_{\mathbf{a}}.$$

The exact fiber theorem implies $\pi_G(x) \in \mathcal{F}(\mathbf{a})$, proving existence.

Assertion (ii) is the intersection of (4.5) with K_G . For (iii), a nonzero element of $I_{\mathbf{a}} \cap K_G$ has a valuation $g \in G$ strictly above every $2d_n$. Conversely every such g is realized by $\omega^{-g} \in K_G$. This is exactly the failure of cofinality. $\square$

Remark 5.2 (The projection is not a homomorphism). The map π_G in the proof is an additive support projection, not a field map. For $G = \mathbb{Z}$, one has $\pi_G(\omega^{1/2}) = 0$ but $\pi_G((\omega^{1/2})^2) = \omega$. The argument uses only the support of the difference from an actual continuant; it never projects a product or a reciprocal.

Remark 5.3 (Why eventual constancy was separated). It would be invalid to demand a k with $d_k > d_n$ in the first case. Ordinary digits improve real coefficient precision without changing valuation. The real continued-fraction theorem is exactly the missing input in that case. The two-case proof also avoids an unjustified summation of the differences $r_{n+1} - r_n$.

Corollary 5.4 (Countable cofinality threshold). *For a nonzero set-sized additive subgroup G of $\mathbf{No}$, some admissible infinite code in $\mathbf{Oz}_G$ has a unique realization in K_G if and only if $G_{\geq 0}$ has a countable cofinal subset.*

Proof. A uniquely realizing code supplies the countable cofinal sequence $(2d_n)$. Conversely choose a nondecreasing cofinal sequence $g_n \in G_{\geq 0}$, with $g_0 = 0$. Set

$$a_0 = 0, \quad a_n = \omega^{g_n - g_{n-1}} \quad (n \geq 1).$$

Zero increments give the digit 1; positive increments give a positive infinite omnific integer. The degree sequence is exactly g_n . Theorem 5.1 gives a unique realization. $\square$

The cofinality statements using (d_n) and $(2d_n)$ are equivalent. One direction is immediate; for the other, apply cofinality of $(2d_n)$ to $2g$ to obtain $d_n \geq g$. This does not divide an element of G by two.

Corollary 5.5 (Convergence in the unique Hahn case). *If $G \neq \{0\}$ and the conditions for uniqueness in Theorem 5.1 hold, the continuants converge in the order topology of K_G to that unique point.*

Proof. For any positive $b \in K_G$, choose n_0 with $d_{n_0} + d_{n_0+1} > v(b)$. Such a strict bound exists by cofinality and a positive element of G . For $n \geq n_0$, (4.3) gives $|x - r_n| < b$, hence $|x - r_n| < b$. $\square$

When $G = \{0\}$, the field is $\mathbb{R}$ and ordinary real convergence applies instead. When $G \neq \{0\}$ but all late digits are ordinary, ordinary real coefficient convergence does *not* imply order convergence in K_G : nonzero real errors exceed every positive infinitesimal.

6 The algebra of precision ideals

6.1 Realizing an arbitrary countably specified precision cut

Proposition 6.1 (Prescribed degree cut). *Let $(\gamma_n)_{n \geq 0}$ be a nondecreasing sequence of nonnegative surreals with $\gamma_0 = 0$. There is an admissible code with*

$$I_a = \{0\} \cup \{e : v(e) > \gamma_n \text{ for every } n\}.$$

One choice is

$$a_0 = 0, \quad a_n = \omega^{(\gamma_n - \gamma_{n-1})/2} \quad (n \geq 1). \quad (6.1)$$

Proof. Every exponent in (6.1) is nonnegative, so each digit is at least one and belongs to $\mathbf{Oz}$. Formula (4.2) gives $2d_n = \gamma_n$, which proves the assertion. $\square$

Within a prescribed G , this construction requires the displayed half increments to belong to G . This is automatic for divisible G when all $\gamma_n \in G$. Theorem 5.1 itself has no such divisibility requirement.

Theorem 6.2 (Prime precision criterion). *The ideal I_a is prime in $\mathcal{O}$ if and only if condition (P) holds. The same equivalence holds for $I_a \cap K_G$ in $\mathcal{O} \cap K_G$ when the digits lie in K_G . If (P) fails, there is an explicit element whose image in the quotient is nonzero but has square zero.*

Proof. The nonnegative valuations of elements outside I_a form the lower set

$$D = \bigcup_{n \geq 0} [0, 2d_n].$$

Primality is equivalent to closure of D under addition. If (P) holds and $g, h \in D$, choose n bounding both by $2d_n$ and then m with $d_m \geq 2d_n$. It follows that $g + h \leq 4d_n \leq 2d_m$, so $g + h \in D$.

Conversely, additive closure applied to $2d_n + 2d_n$ gives an m with $4d_n \leq 2d_m$, which is (P). If this condition fails, choose n such that $d_m < 2d_n$ for every m . Then $d_n > 0$ and

$$e = \omega^{-2d_n} \notin I_a, \quad e^2 = \omega^{-4d_n} \in I_a.$$

When all digits lie in K_G , these monomials lie there too; thus the same necessity proof works in the restricted ring. The sufficiency proof is unchanged. $\square$

Primality for the full class is an elementwise assertion. Quotient rings are taken inside set-sized Hahn workspaces containing the digits and the elements under discussion. Every displayed witness lies in such a workspace. No object consisting of proper-class cosets is assumed.

Corollary 6.3 (Radical and nilpotency index). *For $0 \neq e \in O$, its class modulo I_a in a set-sized workspace as above is nilpotent exactly when*

$$\text{there is } k \geq 1 \text{ such that } k v(e) > 2d_n \text{ for every } n. \quad (6.2)$$

The least such k is its nilpotency index, with index one meaning the zero class. The radical $\sqrt{I_a}$ is prime.

Proof. The criterion and index are the equality $v(e^k) = kv(e)$ and the definition of the ideal. To prove primality, suppose $(uw)^k \in I_a$. Write $g = v(u)$, $h = v(w)$ and assume $g \leq h$. Then $2kh \geq k(g + h) > 2d_n$ for every n , so $w^{2k} \in I_a$ and $w \in \sqrt{I_a}$. Zero factors are immediate. This is the familiar valuation-ring argument, included to identify the precise cut involved. $\square$

Condition (P) is thus more than a growth convenience. It says that the undetected infinitesimals define a reduced integral-domain quotient, rather than an error calculus with nonzero nilpotent classes. It will also be one of the two exact conditions for algebraic rigidity.

7 Algebraic rigidity over the digit field

7.1 The coefficient field must be specified

Define

$$K_a = \mathbb{Q}(a_0, a_1, a_2, \dots), \quad A_a = \{x \in \mathcal{F}(a) : x \text{ is algebraic over } K_a\}.$$

The field K_a is countable: every element is a rational expression in finitely many members of a countable list. The digits themselves may have enormous or uncountable supports, but this does not change the cardinality of the field *generated by them*. Consequently the set of all elements of **No** algebraic over K_a is countable.

These are statements relative to the specified digit field, not to all omnific coefficients or to the whole ambient surreal field. For an arbitrary code we do not know that A_a is nonempty.

Lemma 7.1 (Algebraic errors cannot be newly invisible). *Let $E \subseteq \mathbf{No}$ be a set-sized subfield. If a nonzero $e \in I_a$ is algebraic over E , then $I_a \cap E$ contains a nonzero element. Hence*

$$I_a \cap E = \{0\} \iff I_a \cap \text{Alg}_{\mathbf{No}}(E) = \{0\},$$

*where $\text{Alg}_{\mathbf{No}}(E)$ is the field of elements of **No** algebraic over E .*

Proof. Put $g = v(e) > 0$ and take a nonzero polynomial relation $\sum_j c_j e^j = 0$ over E . Among the valuations $v(c_j) + jg$ of nonzero terms, the minimum occurs at least twice. Otherwise the uniquely leading term could not cancel. For two minimizing indices $i < j$,

$$(j - i)g = v(c_i) - v(c_j) = v(c_i/c_j).$$

The ratio c_i/c_j is a nonzero element of E . Since $g > 2d_n$ for all n , its valuation $(j-i)g$ has the same strict lower bounds. Thus the ratio belongs to I_a . The reverse implication in the equivalence is immediate from $E \subseteq \text{Alg}_{\mathbf{No}}(E)$. $\square$

Lemma 7.2 (What the digit field can see). *With $D_0 = \deg a_0$ for $a_0 \neq 0$, and $D_0 = 0$ otherwise,*

$$I_a \cap K_a = \{0\} \iff (\mathbf{P}) \text{ and } (\mathbf{V}). \quad (7.1)$$

If either condition fails, a nonzero element of the intersection is explicitly available from the digits and continuants.

Proof. If $(\mathbf{V})$ fails, then $a_0 \neq 0$ and $v(a_0^{-1}) = D_0 > 2d_n$ for every n . Thus $a_0^{-1} \in I_a \cap K_a$. If $(\mathbf{P})$ fails, choose n with $d_m < 2d_n$ for every m . The nonzero element $q_n^{-4} \in K_a$ has valuation $4d_n > 2d_m$ for every m , so it belongs to the intersection.

For the converse assume both conditions. Let $z \in K_a^\times$. Write $z = P/Q$, where P, Q are nonzero evaluations of integer-coefficient polynomials in finitely many digits. They belong to $\mathbf{Oz}$. Choose N large enough to include those digit indices and to satisfy $D_0 \leq 2d_N$. Every degree of an appearing digit is then at most $2d_N$. If M bounds the total degrees of both polynomials, every nonzero monomial in them has growth degree at most $2Md_N$. Finite addition, even with cancellation, cannot increase that bound. Therefore

$$0 \leq \deg P, \deg Q \leq 2Md_N, \quad |v(z)| = |\deg Q - \deg P| \leq 2Md_N.$$

If $d_N = 0$, this gives $v(z) = 0$, already excluding $z \in I_a$. Otherwise iterating $(\mathbf{P})$ produces an m with $d_m \geq Md_N$. It follows that $v(z) \leq 2d_m$, again excluding membership. This proves (7.1). $\square$

Theorem 7.3 (Exact algebraic-fiber trichotomy). *For every admissible code the following statements hold.*

- (i) *If $(\mathbf{P})$ and $(\mathbf{V})$ both hold, then $|A_a| \leq 1$.*
- (ii) *If either condition fails, then A_a is either empty or countably infinite.*
- (iii) *In the nonempty case of (ii), for any $\xi \in A_a$ there is $0 \neq \delta \in I_a \cap K_a$ such that*

$$\xi + q\delta \in A_a, \quad K_a(\xi + q\delta) = K_a(\xi) \quad (q \in \mathbb{Q}). \quad (7.2)$$

Thus whenever an algebraic realization exists, it is unique if and only if both displayed conditions hold.

Proof. If ξ, η are algebraic realizations, their difference is algebraic over K_a and belongs to I_a by Theorem 4.4. Lemmas 7.1 and 7.2 force that difference to vanish when both conditions hold.

When either fails, Lemma 7.2 supplies a nonzero δ in the digit field and precision ideal. Given one algebraic ξ , the exact fiber theorem and $\mathbb{R}$ -linearity of the ideal give all the points in (7.2). Translation by an element of the base field does not change the generated extension. The points are distinct as q varies. Countability of the entire algebraic closure inside $\mathbf{No}$ gives the upper bound and hence the exact countable cardinality. $\square$

7.2 Two different sources of failure

Condition $(\mathbf{P})$ is intrinsic to the precision ideal: the observed denominator degrees must eventually absorb their own doubles. Condition $(\mathbf{V})$ is different. The first digit contributes a parameter to the coefficient field but contributes nothing to any denominator degree. It may therefore introduce a scale invisible to all subsequent digits.

This distinction explains why even a long periodic tail can fail to give algebraic uniqueness. It also explains why the full-class nonuniqueness of Corollary 4.5 is compatible with uniqueness among algebraic elements over a countable coefficient field.

8 Periodic itineraries and periodic complete quotients

8.1 A distinguished algebraic realization always exists

Assume an eventually periodic code is displayed as

$$\mathbf{a} = (a_0, \dots, a_{r-1}, \overline{b_1, \dots, b_\ell}), \quad r \geq 1, \quad \ell \geq 1, \quad (8.1)$$

where the b_j are positive omnific integers. The finite digit field is

$$K = \mathbb{Q}(a_0, \dots, a_{r-1}, b_1, \dots, b_\ell) = K_{\mathbf{a}}.$$

Let

$$M = \prod_{j=1}^{\ell} \begin{pmatrix} b_j & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad T(t) = \frac{At + B}{Ct + D}. \quad (8.2)$$

Here $B, C > 0$, $A, D \geq 0$, and $\det M = (-1)^\ell$.

Theorem 8.1 (Period-fixed representative). *There is exactly one $t > 1$ satisfying $T(t) = t$. It is the positive root of*

$$Ct^2 + (D - A)t - B = 0, \quad (8.3)$$

namely

$$t = \frac{A - D + \sqrt{(D - A)^2 + 4BC}}{2C}. \quad (8.4)$$

Its complete quotients repeat after the block. The point $\xi = \Phi_{r-1}(t)$ belongs to $\mathcal{F}(\mathbf{a})$ and satisfies $[K(\xi) : K] \leq 2$. It is the unique point in that fiber whose complete quotient at index r is fixed by this period transformation.

Proof. The discriminant is positive and $\mathbf{No}$ is real closed. The two roots have product $-B/C < 0$, so exactly one is positive. Moreover $T(1) > 1$: a finite composition of the maps $u \mapsto b_j + 1/u$, applied at $u = 1$, has first value strictly greater than $b_1 \geq 1$. Thus the polynomial in (8.3) is negative at 1, and its positive root exceeds 1.

For this root, the identity $t = T(t)$ presents one full block of digits with a final tail equal to $t > 1$. All intermediate tails are greater than one, so the floor algorithm produces the block and returns to the same complete quotient. Induction repeats this indefinitely. Applying the finite prefix gives ξ with the code (8.1). The quadratic equation and the fractional linear prefix place ξ in an extension of degree at most two. Finally, the prefix map is injective, and (8.3) has only one root greater than one. $\square$

This is the familiar fixed-point mechanism behind periodic continued fractions, extended by ordered-field algebra. It is not the assertion that *every* point with periodic digits satisfies (8.3). A periodic itinerary does not force its complete quotients to recur as equal numbers.

8.2 The exact algebraic dichotomy

Define

$$E = \sum_{j=1}^{\ell} \deg b_j, \quad (8.5)$$

$$D_* = \max(\{0, D_0\} \cup \{\deg a_j : 1 \leq j < r\} \cup \{\deg b_j : 1 \leq j \leq \ell\}).$$

Thus E is the degree gained per full period. It is not the degree of a matrix entry chosen without regard to the recurrence.

Theorem 8.2 (One or countably many algebraic periodic realizations). *For the code (8.1), exactly one point of $\mathcal{F}(\mathbf{a})$ is algebraic over K if and only if one of the following holds:*

$$E = D_* = 0 \quad \text{or} \quad E > 0 \text{ and } D_* \leq NE \text{ for some ordinary } N \geq 1. \quad (8.6)$$

In that case the point is ξ from Theorem 8.1. Otherwise there are countably infinitely many algebraic realizations. In particular there are infinitely many of degree at most two, all lying in the single extension $K(\xi)$.

Proof. Put $S = \sum_{j=1}^{r-1} \deg a_j$ and $E_t = \sum_{j=1}^t \deg b_j$, with $E_0 = 0$. For $k \geq 0$ and $0 \leq t < \ell$,

$$d_{r-1+k\ell+t} = S + kE + E_t. \quad (8.7)$$

If $E = D_* = 0$, both conditions (P) and (V) hold. If $E > 0$ and $D_* \leq NE$, all prefix degrees and all partial-period sums are bounded by ordinary multiples of E . The unbounded ordinary multiples kE in (8.7) then absorb both the double of any specified d_n and D_0 . Again both conditions hold. Theorem 7.3 gives at most one algebraic point and Theorem 8.1 supplies it.

Suppose (8.6) fails. Then $D_* > 0$, and either $E = 0$ or $D_* > NE$ for every ordinary N . Choose a displayed nonzero digit b with degree D_* , allowing $b = a_0$ if necessary, and put

$$\delta = b^{-(2r+2)} \in K^\times.$$

Its valuation is $(2r+2)D_*$. The prefix bound $S \leq (r-1)D_*$ and (8.7) show that this valuation exceeds $2d_n$ for every n . For $E > 0$, use $2(k+1)E < D_*$ for each ordinary k ; for $E = 0$ the assertion is immediate. Thus $\delta \in I_a \cap K$. The points $\xi + q\delta$, $q \in \mathbb{Q}$, are distinct algebraic realizations and generate exactly $K(\xi)$. Countability gives the asserted total cardinality of algebraic points. $\square$

Corollary 8.3 (Pure periodicity has one algebraic representative). *For a purely periodic admissible code with positive first digit, exactly one realization is algebraic over the field generated by its period block. All other realizations are transcendental over that field.*

Proof. Display the code with a one-digit prefix and a cyclically rotated block. Every displayed digit occurs in the period, so $D_* \leq E$ when $E > 0$. If $E = 0$, all digits are ordinary and $D_* = 0$. Apply Theorem 8.2. $\square$

8.3 A stricter cofinality threshold for periodic codes

A positive element e of an ordered abelian group G is an *order unit* if every $g \in G_{\geq 0}$ is bounded above by an ordinary multiple of e .

Theorem 8.4 (Periodic uniqueness detects an order unit). *Let $G \neq \{0\}$ and let (8.1) be a code in $\mathbf{Oz}_G$. Its realization in K_G is unique if and only if its period degree E is a positive order unit of G . Consequently:*

$$\begin{aligned} \text{some infinite code is unique in } K_G & \iff G \text{ has countable cofinality,} \\ \text{some eventually periodic code is unique in } K_G & \iff G \text{ has a positive order unit.} \end{aligned}$$

Proof. If E is an order unit, the terms $S + kE$ in (8.7) are cofinal in $G_{\geq 0}$. Theorem 5.1 gives uniqueness.

Conversely, uniqueness makes (d_n) cofinal. It cannot happen with $E = 0$, because then the degrees are eventually bounded by S . Given $g \in G_{\geq 0}$, choose a sufficiently late $n = r-1+k\ell+t$ with $d_n \geq S+g$. Formula (8.7) implies $g \leq kE + E_t \leq (k+1)E$. Thus E is an order unit.

The first existence equivalence is Corollary 5.4. For the second, an order unit e gives the constant code $(\omega^e, \omega^e, \dots)$, whose degree sequence is $d_n = ne$. Necessity follows from the preceding paragraph. $\square$

This separates two notions often conflated by rank-one intuition. A value group can have countable cofinality without any order unit. In such a field some infinite codes determine unique points, but no eventually periodic code does so.

9 Worked examples and boundary cases

9.1 The golden ratio and its whole infinitesimal neighborhood

For $\mathbf{a} = (1, 1, 1, \dots)$ all $d_n = 0$. Hence

$$\mathcal{F}(\mathbf{a}) = \varphi + \mathfrak{m}, \quad \varphi = \frac{1 + \sqrt{5}}{2}. \quad (9.1)$$

The algebraic representative over $K = \mathbb{Q}$ is uniquely φ . For example, $\varphi + \omega^{-1}$ has exactly the same infinite digits but is not algebraic over $\mathbb{Q}$. Indeed it is a nonreal surreal, and no nonreal surreal is algebraic over the real-closed ordered subfield $\mathbb{R}$. The example illustrates the older nonuniqueness phenomenon; the new classification explains its algebraic rigidity at the same time.

In $K_{\{0\}} = \mathbb{R}$ the fiber is a singleton. In every K_G with $G \neq \{0\}$ it is not, since K_G has a positive infinitesimal. Ordinary real continued-fraction uniqueness is therefore exactly consistent with (9.1).

9.2 An initial scale that a periodic tail never sees

For $\mathbf{a} = (\omega, 1, 1, 1, \dots)$ again $I_{\mathbf{a}} = \mathfrak{m}$, but now $K = \mathbb{Q}(\omega)$. A period-fixed realization is

$$\xi = \omega + \frac{1}{\varphi} = \omega + \varphi - 1.$$

Every $\xi + q/\omega$, $q \in \mathbb{Q}$, has the same code and is algebraic of degree two over K . Here (P) holds, whereas (V) fails: $D_0 = 1$ but every $d_n = 0$. Thus even primeness of the precision ideal does not imply algebraic uniqueness over the digit field.

9.3 One infinite later digit and a nonreduced error quotient

Take $\mathbf{a} = (0, \omega, 1, 1, \dots)$. Then $d_0 = 0$ and $d_n = 1$ for $n \geq 1$, so

$$I_{\mathbf{a}} = \{0\} \cup \{e : v(e) > 2\}.$$

The element ω^{-2} is outside the ideal but its square is inside. The quotient therefore has a nonzero square-zero class. Here (V) holds and (P) fails. A distinguished point is $\xi = 1/(\omega + 1/\varphi)$, and $\xi + q\omega^{-3}$ gives infinitely many algebraic realizations over $\mathbb{Q}(\omega)$.

9.4 A constant infinite period

For $\mathbf{a} = (\omega, \omega, \omega, \dots)$ the positive fixed point is

$$\begin{aligned} \rho &= \frac{\omega + \sqrt{\omega^2 + 4}}{2} \\ &= \omega + \omega^{-1} - \omega^{-3} + 2\omega^{-5} - 5\omega^{-7} + 14\omega^{-9} - \dots \end{aligned} \quad (9.2)$$

The binomial series at the infinitesimal ω^{-2} gives the displayed Hahn expansion. More precisely its coefficient of ω^{-2k-1} is $(-1)^k C_k$, where $C_k = \frac{1}{k+1} \binom{2k}{k}$. Here $d_n = n$, and

$$I_{\mathbf{a}} = \{0\} \cup \{e : v(e) > 2n \text{ for every ordinary } n\}.$$

There is a unique realization in $\mathbb{R}((\omega^{\mathbb{Q}}))$ and in $\mathbb{R}((\omega^{\mathbb{R}}))$, but not in **No**: the perturbation $\omega^{-\omega}$ is invisible to the entire code. Nevertheless ρ is the unique realization algebraic over $\mathbb{Q}(\omega)$.

The code is periodic, its full-class fiber is large, its rank-one Hahn fiber is a singleton, and its digit-field algebraic fiber is a singleton. These four statements concern different ambient categories and do not contradict one another.

9.5 A bounded increasing cut, with a genuinely different boundary

Let $a_0 = 0$ and $a_n = \omega^{2^{-n}}$ for $n \geq 1$. Then

$$d_n = 1 - 2^{-n}, \quad I_a = \{0\} \cup \{e : v(e) > 2 - 2^{1-n} \text{ for every } n\}.$$

In the rank-one field $K_{\mathbb{R}}$, this becomes

$$I_a \cap K_{\mathbb{R}} = \{0\} \cup \{e \in K_{\mathbb{R}} : v(e) \geq 2\}. \quad (9.3)$$

The endpoint is included. In the full surreal value group one must retain the sequence of inequalities: the value $2 - \omega^{-1}$ satisfies them all but is less than 2. Replacing a real supremum by a nonexistent surreal supremum would give the wrong ideal.

Theorem 5.1 ensures a realization in $K_{\mathbb{R}}$ even though the cumulative degrees are bounded and the fiber is not a singleton. This example exercises the non-eventually-constant support-projection case, not the ordinary-tail case.

9.6 Countable cofinality without an order unit

Let

$$G = \bigoplus_{j \geq 1} \mathbb{Q} \omega^j \subseteq (\mathbf{No}, +),$$

where the direct sum means finite rational linear combinations. This ordered group has countable cofinality, witnessed by the sequence ω^j , but has no order unit: a finite linear combination is dominated by a sufficiently higher power of ω .

The code $a_0 = 0$, $a_n = \omega^{\omega^n}$ has cumulative degrees $d_n = \sum_{j=1}^n \omega^j$, cofinal in G . It therefore has a unique realization in K_G . On the other hand, Theorem 8.4 excludes uniqueness for every eventually periodic code in this field. This is an explicit infinite-rank separation between the two uniqueness thresholds.

10 Arbitrarily large algebraically independent families in a fiber

The size of a full surreal fiber is not merely a counting defect. Its undetected directions can carry arbitrarily much algebraic information. The following statement is deliberately set-indexed, so it does not presuppose a proper-class transcendence basis.

Theorem 10.1 (Hidden algebraic independence). *Let $\mathbf{a}$ be any admissible infinite code, let $E \subseteq \mathbf{No}$ be a set-sized subfield, and let κ be any set cardinal. Then $\mathcal{F}(\mathbf{a})$ contains a family of κ elements algebraically independent over E .*

Proof. Choose $\xi \in \mathcal{F}(\mathbf{a})$. Let H be the rational span of all growth exponents occurring in the normal forms of ξ and all members of E . This is a set-sized subgroup of **No**, and $E \cup \{\xi\} \subseteq \mathbb{R}((\omega^H))$. Choose an ordinal β so large that

$$|h| < \omega^\beta \quad (h \in H), \quad 2d_n < \omega^\beta \quad (n \geq 0). \quad (10.1)$$

Such a β exists because the quantities to be bounded form a set and ordinals are cofinal in $\mathbf{No}$.

For $\alpha < \kappa$ put

$$\theta_\alpha = \omega^{\beta+\alpha+1}, \quad \delta_\alpha = \omega^{-\theta_\alpha}.$$

Each δ_α belongs to I_a by (10.1). Moreover the family (θ_α) is rationally independent modulo H : a nonzero finite rational linear combination has a largest growth exponent at least $\beta + 1$, and hence absolute magnitude greater than ω^β . It cannot lie in H .

Consider a nonzero polynomial in finitely many δ_α over $\mathbb{R}((\omega^H))$. A monomial with multi-index ν shifts the support of its nonzero coefficient into the coset

$$H - \sum_\alpha \nu_\alpha \theta_\alpha.$$

Distinct multi-indices give distinct cosets by the independence just proved. The nonzero polynomial summands consequently have disjoint supports and cannot cancel. Thus the δ_α are algebraically independent over $\mathbb{R}((\omega^H))$.

Finally set $x_\alpha = \xi + \delta_\alpha$. All these points lie in the fiber. Translation of indeterminates by ξ , which belongs to the coefficient Hahn field, preserves nonzero polynomials. The x_α are therefore algebraically independent over E as claimed. $\square$

Remark 10.2 (What this does and does not imply). The theorem holds for each prescribed set field E and each prescribed set cardinal separately. It does not choose one family independent over *all* set subfields simultaneously. Such a reading would be false, since a field could contain a chosen member. Nor does it claim a first-order definable or algorithmically computable independent family.

11 Surcomplex fibers and changes of coordinate

11.1 Coordinatewise coding

There is no order-compatible floor on $\mathbf{No}[i]$. We therefore specify the construction rather than silently importing real continued fractions. For $z = x + iy$, record the omnific floor code of x and the omnific floor code of y , whenever both algorithms are nonterminating.

Theorem 11.1 (Exact surcomplex coordinate fiber). *If $x \in \mathcal{F}(a)$ and $y \in \mathcal{F}(b)$, then the points with the same pair of coordinate codes form exactly*

$$(x + iy) + (I_a + iI_b). \tag{11.1}$$

If $I_a = I_b = I$, then $I + iI$ is an ideal of $O_{\mathbb{C}} = O + iO$. With

$$v_{\mathbb{C}}(u + iw) = \min\{v(u), v(w)\},$$

it is the corresponding surcomplex valuation ideal. If the two ideals are different, their sum in (11.1) is not invariant under multiplication by i .

Proof. The exact real fiber theorem applied to both coordinates proves (11.1). If the ideals agree, then

$$(a + ib)(e + if) = (ae - bf) + i(af + be) \in I + iI \quad (a, b \in O, e, f \in I).$$

The complex normal form combines the real and imaginary coefficients; a nonzero real coefficient and a nonzero imaginary coefficient cannot cancel each other. This proves the displayed valuation formula and the valuation-ideal assertion. If multiplication by i preserved $I_a + iI_b$, applying it to real and imaginary elements would force both inclusions $I_a \subseteq I_b$ and $I_b \subseteq I_a$. $\square$

Equal cuts therefore give a cut-defined intersection of valuation balls, whereas unequal cuts give an anisotropic coordinate neighborhood. This is a coordinatewise theory, not a theorem about nearest-Gaussian-integer continued fractions or a rotation-invariant coding of all surcomplex points.

Corollary 11.2 (Surcomplex hidden independence). *Given a pair of admissible infinite coordinate codes, a set-sized subfield $E \subseteq \mathbf{No}[i]$, and any set cardinal κ , their common coordinate fiber contains κ points algebraically independent over E .*

Proof. Fix $z = x + iy$ in the fiber. Gather the growth supports of the real and imaginary parts of z and of all members of E into a rational span H . Use the construction of Theorem 10.1, bounding the degree cuts of both coordinate codes, and work over $\mathbb{C}((\omega^H))$ instead of $\mathbb{R}((\omega^H))$. The real perturbations δ_α are invisible in the real coordinate, and the imaginary coordinate is unchanged. Distinct support cosets again prevent cancellation, including complex coefficients. $\square$

11.2 Exact transport under a fractional linear map

Proposition 11.3 (Precision transport). *Let $f(z) = (az + b)/(cz + d)$ over $\mathbf{No}[i]$, with $ad - bc \neq 0$ and $cz + d \neq 0$. For a nonzero perturbation e satisfying*

$$v_{\mathbb{C}}(ce) > v_{\mathbb{C}}(cz + d), \quad (11.2)$$

one has $c(z + e) + d \neq 0$ and

$$f(z + e) - f(z) = \frac{(ad - bc)e}{(cz + d)(c(z + e) + d)}, \quad (11.3)$$

$$v_{\mathbb{C}}(f(z + e) - f(z)) = v_{\mathbb{C}}(ad - bc) + v_{\mathbb{C}}(e) - 2v_{\mathbb{C}}(cz + d). \quad (11.4)$$

The hypothesis is automatic when $c = 0$, with $v_{\mathbb{C}}(0) = +\infty$.

Proof. The strict inequality makes $c(z + e) + d$ have the same leading valuation as $cz + d$. Subtracting the two fractions proves (11.3); taking valuations proves (11.4). $\square$

Thus within a sufficiently small fiber neighborhood, a projective change of chart translates the valuation threshold by an exact amount. Without (11.2), a pole or leading cancellation can invalidate that linear threshold law. No claim is made that an arbitrary fractional linear map preserves the digit codes themselves.

12 Proof dependencies and reproducible checks

The mathematical dependency chain is short. Normal forms and the omnific floor give the finite matrix identities and cylinders. The two-step buffer and Fibonacci growth give the exact fiber. The exact error valuation then gives full Hahn-field realization by support projection. Independently, the valuation cut gives the prime criterion. Polynomial degree bounds and the elementary leading-term cancellation lemma give the digit-field algebraic dichotomy. Period matrices supply algebraic existence, and support cosets supply hidden independence.

Only the normal-form field construction, the cut property, real closedness of $\mathbf{No}$, and classical real continued-fraction existence are imported. In particular the proof of algebraic rigidity does not require a real-closedness theorem for every Hahn field, a valuation-extension structure theorem, a saturation theorem, or any of the repository's advanced refereed results.

12.1 The finite verification artifact

The supplied program code/`verify.py` uses exact rational arithmetic, finite generalized polynomials with rational growth exponents, and SymPy symbolic identities. The recorded run passed **81,256 assertions**. The main categories are as follows.

Check family	Assertions
Fibonacci denominator inequalities	28,268
Two-step cylinder buffers	4,294
Generalized-polynomial degree laws	1,260
Generalized-polynomial determinant identities	1,260
Rational cylinder widths	5,694
Rational digit recovery	5,694

Other checks cover both endpoint-distance identities, monotonicity, nesting, period matrices, the fractional linear difference formula, a counterexample to multiplicativity of support projection, and the Catalan Laurent expansion through its justified truncation order. The full category counts, fixed seed, Python version, and SymPy version are in `audits/verification.json`.

These are finite consistency checks of identities used by the proof. They cannot check the proper-class existence quantifier, arbitrary infinite supports, countable cofinality, or the absence of every polynomial relation. Those conclusions rest on the written proofs, not on the number of test cases.

12.2 A practical formalization route

A future Lean development should first isolate the ordered-field continuant package from the surreal-specific normal-form package. The matrix identities, cylinder inequalities, and two-step buffer need only finite recursion, ordered-field arithmetic, and discreteness of the digit ring. The exact-fiber proof then adds an Archimedean comparison relation quantifying over ordinary natural numbers and the denominator-degree law.

The next genuinely surreal step is the support projection lemma, stated as an additive subseries construction and explicitly not as a ring homomorphism. The algebraic-error lemma should be proved separately for a valued field, by the repeated-minimum argument in Lemma 7.1. This keeps the main algebraic classification independent of heavy real-closed-field automation.

No Lean file is included as though it had been checked. The repository was not built during this task. A written proof, a finite test run, and a kernel-checked theorem remain three different levels of evidence.

13 Further research questions

The following are proposed problems arising from the proved statements. They are not all asserted to be previously published open problems. The historical comparison is itself part of the research program.

Research question 1 (Priority and exact equivalence of formulations). Compare the valuation-fiber formula, full Hahn realization theorem, and digit-field algebraic criterion against the wider literature following [3]. Which statements are genuinely new, and which are hidden consequences of earlier associated-class or generalized-fraction formulations? A useful outcome would be a theorem-by-theorem concordance, not merely a list of citations.

Research question 2 (Proper subfields of a full Hahn field). Characterize subfields $L \subsetneq K_G$ containing $\mathbb{R}$ and closed under omnific floor for which every admissible code in $L \cap \mathbf{Oz}$ has a realization in L .

Full Hahn fields have the property by Theorem 5.1; arbitrary subfields do not follow from its proof. Determine which support-completeness or truncation assumptions suffice.

Research question 3 (Algebraic existence for a nonperiodic code). Theorem 7.3 determines the size of the algebraic fiber once its nonemptiness is known. Find checkable conditions on nonperiodic omnific digits that imply, or rule out, algebraic existence over the countable digit field. In particular, investigate codes with cofinal degrees and a finitely described recurrence among their digits.

Research question 4 (A correctly scoped Lagrange converse). For a specified full Hahn field and a specified coefficient subfield, which algebraic points of degree two have eventually periodic omnific floor itineraries? The coefficient field and the allowed digits must be fixed in advance. Theorem 8.1 proves an existence direction for a given periodic code, not this converse.

Research question 5 (Simplicity versus dynamical selection). When does the simplest cylinder separator $\{L \mid R\}$ coincide with the period-fixed algebraic representative of an eventually periodic code? How does this comparison behave under finite prefix maps? Establishing precise hypotheses would connect the present valuation classification with the simplicity-based selections in earlier work. Roggeman's residue and periodicity theorems must be compared before any instance is classified as a previously unresolved problem.

Research question 6 (Iteration beyond the first block of length ω). Design and compare limit-stage remainder rules that refine a full fiber without discarding the original point. The transfinite problem is not to redeclare the first-block cut to be its unique value. It is to retain and encode an element of the precision ideal, compatibly with earlier generalized continued-fraction constructions.

Research question 7 (Other integer parts). Which parts of the classification remain true for a different discretely ordered integer part of an ordered field? The finite cylinder and buffer lemmas are robust, but the degree identity, the ordinary-tail argument, and the support projection use special features of $\mathbf{Oz}$. An axiomatic characterization of exactly those features would be valuable.

Research question 8 (Primary precision ideals). Theorems 6.2 and 6.3 classify primality and the radical directly from the degree cut. Characterize when a precision ideal is primary in its valuation ring, and describe the torsion and annihilators in the associated error quotient at higher value-group rank. The rank-one examples suggest useful test cases but do not decide the higher-rank problem.

Research question 9 (A genuinely surcomplex continued-fraction algorithm). Choose an explicit nearest-Gaussian-omnific-integer rule and determine whether any analogue of the exact fiber theorem survives. A central obstacle is that complex denominators can have leading cancellation, so positivity cannot supply either monotonicity or the degree recurrence. The coordinatewise result of Section 11 does not solve this.

Research question 10 (Effective precision and representations). For computably presented exponent groups and exact digit notations, determine the complexity of finding an index at which a specified error becomes visible. Separate the problem of deciding equality of digits from the cofinality problem for their cumulative degrees. A finite program for rational exponents is not an oracle for arbitrary surreal descriptions.

Research question 11 (Birthday bounds). Bound the birthday of a simplest realization from the birthdays of its digits and their support cuts. Compare those bounds with the birthdays of the explicit quadratic representative in the periodic case. Ordinary field degree and surreal birthday should not be identified, and no general product-birthday conjecture is resolved by the present work.

Research question 12 (Formal proof and reusable interfaces). Formalize the buffer lemma, exact fiber theorem, support-projection existence argument, and algebraic dichotomy in the repository's actual surreal field. The decisive interface question is how to expose normal-form subseries and their leading valuations without introducing unproved summability or class-size assumptions.

14 Conclusion

An omnific continued fraction is most naturally treated as precision data. The exact data are the coset $x + I_a$, with an explicit lower cut on allowed error valuations. From that viewpoint three seemingly conflicting phenomena become compatible: every full surreal infinite fiber is large; a full Hahn workspace can make a code unique; and a countable digit field can single out one algebraic member of a much larger fiber.

The two most useful thresholds are different. General uniqueness in a full Hahn field detects countable cofinality of its value group, whereas periodic uniqueness detects an order unit. Algebraic uniqueness over the digit field is controlled instead by self-doubling of the cumulative degrees and visibility of the initial digit. These precise distinctions, rather than the already known failure of ordinary uniqueness, are the substantive outcome of this investigation.

A Notation and scope checklist

Notation	Meaning
No , Oz , No [i]	Surreal field, omnific integer ring, and No [i].
$\deg x$, $v(x)$	Greatest growth exponent and its negative; $v(0) = +\infty$.
O , $\mathfrak{m}$	Finite surreals and infinitesimals.
K_G	Full real-coefficient Hahn field with growth exponents in the set group G .
a , $\mathcal{F}(a)$	An ordinary ω -indexed admissible code and all its nonterminating realizations.
p_n , q_n	Finite continuants; $q_0 = 1$.
r_n , s_n , C_n	Convergent, other cylinder endpoint, and open prefix cylinder.
d_n	$\deg q_n = \sum_{j=1}^n \deg a_j$; the first digit does not contribute.
I_a	Errors whose valuations exceed $2d_n$ for every ordinary n .
K_a	Countable field generated over $\mathbb{Q}$ by the digits themselves.
D_0	Degree of the first digit, with value zero assigned if that digit is zero.
E , D_*	Total degree in a period and largest degree among all displayed digits.

All assertions about uniqueness name their ambient field. All assertions about algebraicity name their coefficient field. All arbitrary-support series are Hahn normal forms, not analytic limits. Prime-ideal claims refer to the finite valuation ring, not to the whole field. Finite tests are not formal verification. The proposed research questions are not advertised as newly solved published conjectures.

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